Idea Submission — HackNow 2026

HOT-04 : Guest Complaint Root-Cause Intelligence

Team TechNova

Submission Essentials:

This submission demonstrates Team TechNova's end-to-end business solution design for HOT-04, addressing how hotels can move from fragmented, manual complaint handling to a structured, ServiceNow-powered guest complaint and root-cause intelligence workflow — built for Clarity, Creativity, Feasibility and Impact.

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Team & Problem Context

Team Information

Team Name	TechNova
College/Institution	Ramachandra College of Engineering Eluru
Number of Members	6
Name of the problem statement chosen	HOT-04 — Guest Complaint Root-Cause Intelligence (Hotel / Hospitality)

Problem Statement

State the exact problem:

- Hotels collect guest complaints across many channels — app, phone, email, front desk and chat — but the process is handled manually and inconsistently, with no single system linking a complaint to its origin (housekeeping, facilities, food & beverage, or front desk).
- Information reaching decision-makers is often incomplete or delayed, so recovery actions and root-cause fixes both suffer.
- There is no clear ownership, status visibility or timely escalation, which means the same issue (e.g. AC faults on a specific floor, or slow room-service delivery) can recur for weeks before anyone notices the pattern.

- Stakeholders impacted: guests (experience and trust), front-line and department staff (unclear accountability), department heads and General Managers (no visibility into systemic issues), and hotel brand/ownership (repeat complaints erode CSAT, reviews and revenue).

This matters because unresolved recurring issues are far more damaging to a hotel's reputation than isolated one-off complaints — and today's manual process cannot reliably tell the difference between the two.

Solution Innovation

Your Approach

TechNova's solution is a ServiceNow Customer Service Management (CSM) based Guest Complaint Root-Cause Intelligence platform that captures every complaint as a structured Case, automatically tags it with category, location, service area and severity, and routes it to the right department through Workflow Studio.

- Guests raise complaints through any channel (mobile app, web portal, call centre, front desk or chatbot); a Virtual Agent and Service Portal create a Case automatically.
- Each Case captures category, exact location (room/floor/outlet), service area and guest impact using the Common Service Data Model (CSDM) so complaints link to actual hotel configuration items.
- Workflow Studio routes the Case to the owning department queue, and the case owner decides between an immediate recovery action and a longer corrective-action task.
- Performance Analytics continuously scans for repeat/systemic issues (e.g. 3+ similar complaints in 30 days for the same category and location) and automatically raises the case through a defined escalation tier.
- Every step — communication, action taken, and closure — is logged on the Case, giving a single auditable record from intake to resolution.

Innovation & Differentiation

- Most hotels track complaints in spreadsheets, PMS notes or generic ticketing tools with no automatic root-cause or recurrence detection — TechNova's solution makes recurrence detection a first-class, automated capability rather than a manual reporting exercise.
- Using ServiceNow's CSDM to tie every complaint to a real configuration item (room, floor, outlet, equipment) allows true root-cause analysis by asset and location, not just by complaint category.
- A tiered, SLA-driven escalation model (Front-line → Supervisor → Department Head/GM → Management/Quality Council) removes the ambiguity of "who owns this" that plagues manual processes today.
- Predictive Intelligence auto-suggests likely root-cause categories from complaint text, speeding up triage and reducing miscategorisation.

Why This Matters

- Faster, clearer ownership shortens guest recovery time and improves CSAT and online review scores.
- Automated recurrence detection turns complaint data into a continuous-improvement engine, reducing repeat issues and associated cost (comps, refunds, staff time).
- Management gains real-time visibility into operational hot-spots (by location, category and department) instead of discovering problems only through guest reviews.

Implementation & Feasibility

Technical Stack

Component	Technology / Module
Front-end and back-end elements that will be included	ServiceNow Now Experience Framework (Service Portal / CSM UI) for guest & agent front-end; ServiceNow Glide/Flow Designer server-side logic and REST APIs as the back-end.
ServiceNow modules that will be used	Customer Service Management (CSM), Workflow Studio, Performance Analytics, Predictive Intelligence, Virtual Agent, Integration Hub, Common Service Data Model (CSDM).
ServiceNow features that will be part of this solution	Case Management, Flow Designer automation, SLA definitions, Assignment Rules, Notifications, Dashboards/Reports, Mobile Agent App, Surveys (CSAT).
High level design	Omni-channel intake creates a CSM Case → auto-classification (category, location, service area, impact) → routing to the owning department queue → root-cause triage → immediate recovery + corrective-action task → recurrence detection via Performance Analytics → tiered escalation → guest closure & CSAT → management dashboard.
Supporting Tools/Languages	JavaScript (server & client scripts), Glide APIs, REST/Integration Hub spokes, HTML/CSS for portal widgets, Performance Analytics scripting, Figma (UX mockups).

Project Timeline

- **Phase 1 (Day 1, first half) — Requirements & data model: finalise complaint taxonomy, CSDM configuration items (rooms/floors/outlets), and user journey mapping.**
- Phase 2 (Day 1, second half) — Core build: Case form, intake channels (Portal + Virtual Agent), category/location auto-capture, routing rules in Workflow Studio.
- Phase 3 (Day 2, first half) — Escalation & analytics: SLA definitions, tiered escalation flows, recurrence-detection logic, Performance Analytics dashboards.
- Phase 4 (Day 2, second half) — Integration, CSAT survey, polish and demo rehearsal; final QA and documentation.

Feasibility Assessment

- Realistic for the hackathon timeline: the solution uses out-of-the-box ServiceNow CSM, Workflow Studio and Performance Analytics capabilities, which are configuration-first rather than custom-code-heavy.
- All required skills are covered within the team — solution architecture, process/UX design, Workflow Studio development, analytics and integration, and QA/documentation.
- Key risks: limited demo-instance data for realistic recurrence patterns (mitigated by seeding sample complaint data); integration with real hotel backend systems is mocked for the hackathon and flagged as a production dependency.

Impact & Metrics

Measurable Outcomes

- Complaint recurrence rate (by category and location) — target: visible reduction over time as corrective actions take effect.
- Mean time to acknowledge and mean time to recovery (Tier 1 and Tier 2 SLAs).
- Guest CSAT score post-resolution, captured via automated survey on Case closure.
- Percentage of complaints correctly auto-categorised and routed without manual re-assignment.
- Number of systemic issues identified and closed via corrective action per month.

Real-World Applicability

- The solution is built on standard ServiceNow CSM/CSDM constructs, so it can scale from a single property to a multi-property hotel group by extending the CSDM hierarchy (property → floor → room/outlet).
- Path to production: pilot in one property, connect real PMS/Housekeeping/F&B systems via Integration Hub, then roll out group-wide with brand-level dashboards.
- Beyond hospitality, the same root-cause and recurrence-detection pattern applies to facilities management, retail store operations, or any environment with distributed physical locations and service quality issues.

Team Roles & Expertise

Role Assignment

Name	Role	Team Lead
B. Pravallika	Team Lead & ServiceNow Solution Architect — owns end-to-end workflow design, Case/CSDM data model, and integration architecture	Yes
K. Meghana	Process & UX Designer — designs guest journey, Service Portal / Virtual Agent screens and complaint intake forms	No
K. Rahini	ServiceNow Developer (Workflow Studio) — builds Flow Designer automations, routing rules, SLA and escalation logic	No
M. Samatha	Data & Analytics Lead — configures Performance Analytics, recurrence-detection logic and KPI dashboards	No
G. Hemanjali Bhavani	Integration & QA Engineer — builds Integration Hub connectors to PMS/Housekeeping/F&B systems and leads testing	No
S. Dhanush	Presentation & Documentation Lead — coordinates demo storyline, documentation and stakeholder presentation	No

Leadership and Coordination

- Team Leader: B. Pravallika leads solution architecture and coordinates the team across design, build and presentation workstreams.
- Skill Mapping: each member's role above is mapped to their strength — process/UX design, ServiceNow development, data & analytics, integration/QA, and documentation/presentation — ensuring parallel workstreams throughout the hackathon.

Diagrams & Visual Aids

The diagrams below illustrate the end-to-end process flow, the ServiceNow-based solution architecture, and the roles/escalation model for the Guest Complaint Root-Cause Intelligence solution.

1. End-to-End Process Flow

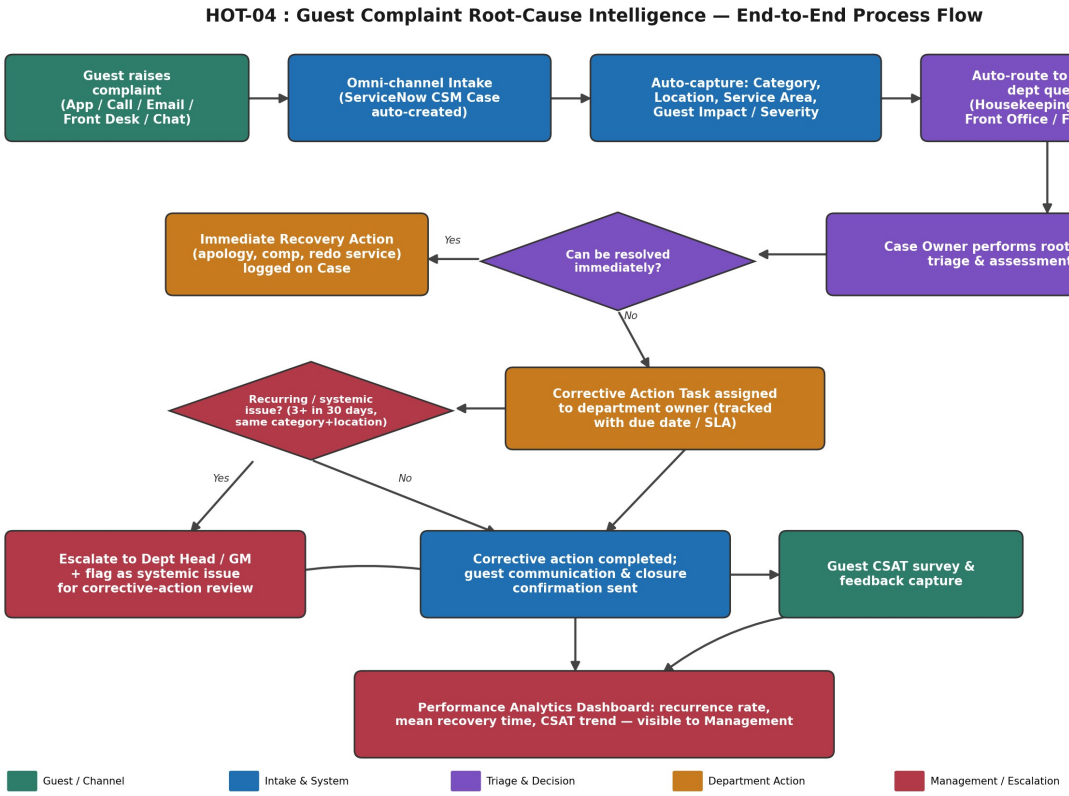


Figure 1 — Guest complaint intake, triage, recovery, escalation and analytics flow

2. Solution Architecture

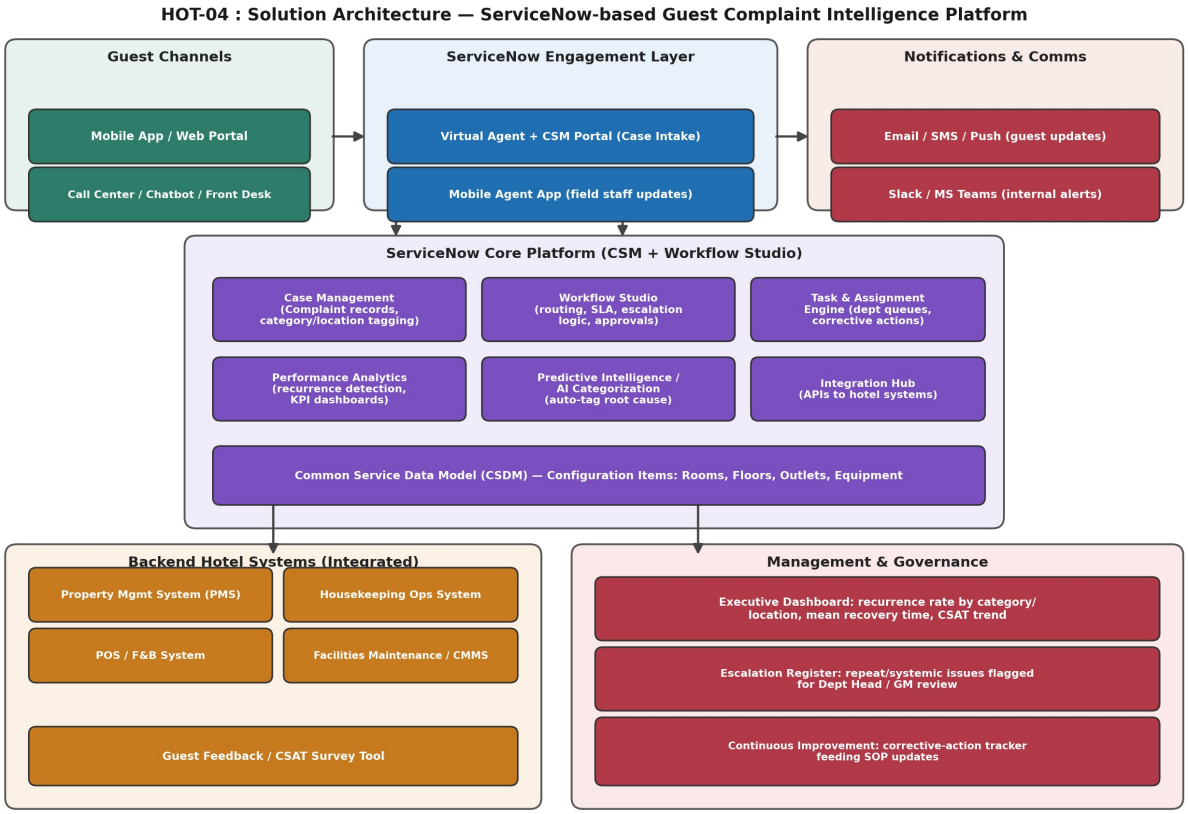


Figure 2 — ServiceNow CSM-based architecture across channels, core platform, backend systems and management layer

3. Roles, Ownership & Escalation Tiers

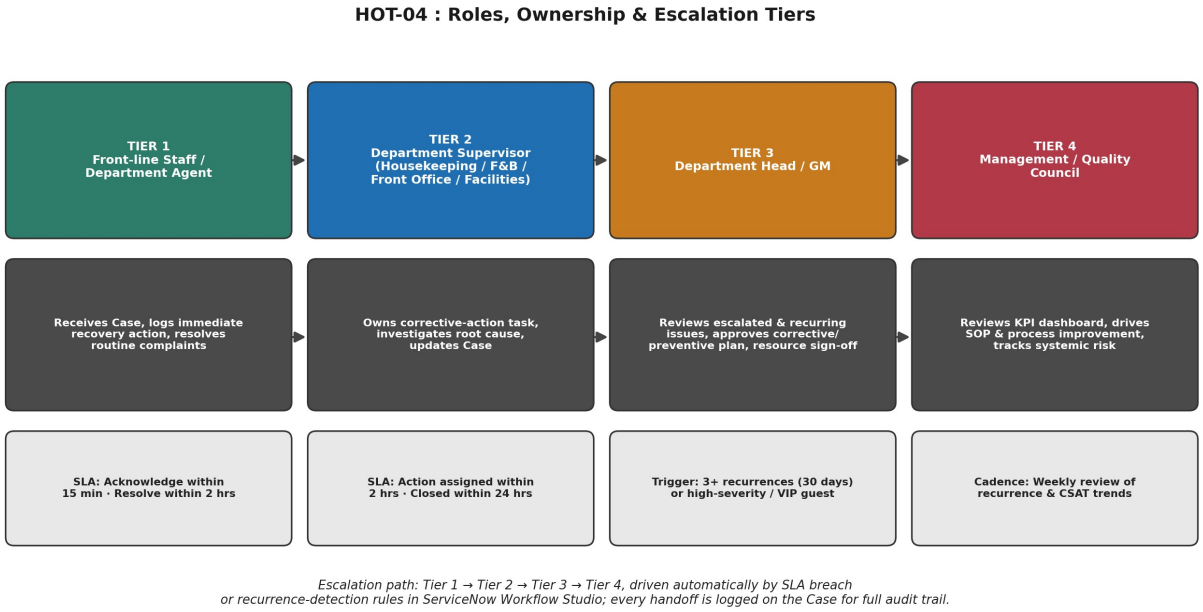


Figure 3 — Four-tier ownership and SLA-driven escalation model

References

- ServiceNow Documentation: https://docs.servicenow.com
- ServiceNow Customer Service Management (CSM) Product Overview
- ServiceNow Common Service Data Model (CSDM) Guidance